TITLE:

Musculoskeletal injury (MSKI) outcome definitions reported in military population research from 2015-2025: A scoping review protocol

AUTHORS:

Jacob A. Naylor, PT, DPT[1,2]

Evan Mayo-Wilson, PhD[1]

Chelsea L. Martin, PT, DPT[1]

Carrie Baldwin-SoRelle, MS[5]

Shabbar Ranapurwala, PhD[1,4]

Michelle Canham Chervak, PhD[3]

Michele Jonsson-Funk, PhD[1]

Stephen W. Marshall, PhD[1,4]

AUTHOR AFFILIATIONS:

[1] Department of Epidemiology, Gillings School of Global Public Health, University of North Carolina, Chapel Hill

[2]Medical Center of Excellence, United States Army

[3]Defense Centers for Public Health-Aberdeen, Defense Health Agency

[4]Injury Prevention Research Center, University of North Carolina, Chapel Hill

[5]Health Sciences Library, University of North Carolina at Chapel Hill

Corresponding Author:

Jacob Naylor, PT, DPT; Email: naylorja@unc.edu

Disclosure:

This review protocol and all associated parts do not represent the official position of the Defense Health Agency, Department of the Army, Department of Defense, or any agency in the U.S. Government.

ABSTRACT

Background: Musculoskeletal injury (MSKI) is a problem within militaries worldwide. There is heterogeneity in MSKI outcome definitions, including what constitutes incident injury, what comprises estimates of MSKI occurrence, and whether time-loss is included, etc. Inconsistent measurement impedes synthesis and hinders the ability to identify the most effective interventions to reduce MSKI.

Objective: This scoping review aims to describe the ways in which MSKI has been defined in research conducted in military populations from 2015 to present.

Methods: A literature search for manuscripts related to musculoskeletal injury in military populations will be developed with a health sciences librarian and conducted in PubMed, Embase, Scopus, CINAHL, SPORTdiscus and Defense Technical Information Center (DTIC) databases. Included reports will be published in English from 2015, the year ICD-10 diagnosis codes were implemented in United States military health records. From full-text articles meeting inclusion criteria, two authors will independently extract the case definitions of MSKI, whether incident/reinjury/exacerbation MSKI were estimated separately, if prior injury history was measured, whether and how estimates of time-loss were reported, and the data source(s) used for each of these. We will report the review following the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping reviews (PRISMA-SCR).

Ethics and Dissemination: No ethics approval is needed for this scoping review, as it relies solely on literature that is already published. This review will be disseminated as a published manuscript in a peer-reviewed journal, at relevant scientific conferences and through other media as appropriate.

Key words: military, musculoskeletal injury, scoping review

Introduction

Musculoskeletal injury (MSKI) hinders military readiness world-wide[1]. In a recent report from the United States (US), MSKI represented 81% of all injuries in active duty personnel [2]. That same year MSKI was responsible for the time-loss of 8.8 million limited duty days in the US Army alone [3]. Globally, a variety of recent analyses have reported MSKI outcomes in subsets of military populations, such as female British Army basic trainees, Australian infantry personnel, or females in US Army ground combat versus non-ground combat occupations[4–6]. Other research has focused on specific types of MSKI, such as ankle sprains in male Israeli infantry trainees, neck pain in Chinese pilots or anterior cruciate ligament tears in Iranian military personnel [7–9]. Military recruits, one of the most widely researched subpopulations subject to MSKI, were shown in a recent systematic review and meta-analysis to have MSKI incidence rates ranging from 0.62 to 19.52 medical attention MSKI, and 0.75 to 3.97 time-loss MSKI per 1000 training days, respectively [10].

However, despite a wide variety of peer-reviewed MSKI research in military populations, there is heterogeneity of MSKI case definitions and estimated outcomes used across manuscripts. This heterogeneity has been noted in systematic reviews for MSKI case definitions, time-loss characterization and severity classification [1,11–14]. Further, this heterogeneity limits research synthesis and can negatively affect the design and evaluation of MSKI prevention interventions where outcomes have not been clearly defined [15]. This could lead to incorrect interpretation of results and sub-optimal decision making using these data.

Sports injury researchers have sought to address problems of MSKI outcome heterogeneity through consensus statements in sports such as cricket[16], football (soccer),[17] swimming,[18] athletics,[19] and rugby.[20] In 2020, The International Olympic Committee consensus statement by Bahr et al. was published, further urging consistency in the recording and reporting of injury (and illness) in sport.[21] For example, these authors note that not all injury complaints result in medical attention (i.e. evaluation by a medical professional) or time-loss.[21,22] They also highlight the differences between index (first) and subsequent injury, and reinjury (an injury of the same type to the same location from which one had fully recovered) and exacerbation (an injury of the same type to the same location from which one had not yet fully recovered).[21,23] They also discuss time-loss as common measurement of injury severity.[21] Several sport-specific extensions of this paper have since been published.[24–29]

Military researchers have also recently sought consensus related to the collection of MSKI outcome data in research conducted in military populations through two Delphi studies. First, in a Delphi study focused on injury surveillance in special operations forces, 13 of 16 experts concurred injury case definitions should be categorizable into one of the following: all complaints (including non-medical attention or injuries without time-loss), medical attention, time-loss, or performance impairment.[30] The second, a Delphi study on the minimum common data elements for surveillance and Reporting of Musculoskeletal Injuries in the MILitary (ROMMIL), demonstrated agreement on outcome-related data elements such as mechanism of injury, and differentiating between overuse, acute and traumatic injuries; however, this study did not explicitly include case definition as an element for consensus.[15]

The purpose of this review is to assess the scope of MSKI case definitions and associated elements published in military research over the last decade. We will present the range and nature of MSKI case definitions and corresponding estimates of occurrence, and the data sources used to acquire this information. The expected result of this review is to synthesize this information in an accessible manner, generate discussion about best practices in injury epidemiology research in military populations, and make recommendations for future studies.

Methods

A literature search including terms related to MSKI and military will be developed with a health sciences librarian and conducted in PubMed, Scopus, CINAHL, SPORTdiscus and Defense Technical Information Center (DTIC) databases. A sample search strategy for PubMed can be found in Appendix A. Included articles will be English-language manuscripts reporting estimates of MSKI occurrence, with or without intervention, in military populations from 01 January 2015 through present. We selected 2015 as the start date because the United States military formally adopted ICD-10-CM diagnoses code use in the electronic medical record and expanded the potential for an injury taxonomy.[31,32] Systematic reviews, commentaries and other secondary reports will be excluded (see Table 1 for complete inclusion and exclusion criteria). COVIDENCE will be used for all article screening and abstraction[33].

As per published recommendations[34,35], two members of the review team (JAN, CLM) will independently screen each title and abstract for initial inclusion.

Interrater reliability will be assessed in COVIDENCE after 10% of abstracts are screened. Full text manuscripts will be sought for initially included articles. The same reviewers will then read the full text of each included article and make a final determination of inclusion or exclusion. A third member of the review team (EMW or SWM) will determine inclusion or exclusion in the event of disagreement that cannot be resolved.

Inclusion Criteria	Exclusion Criteria
Original research reporting MSKI measures of occurrence without limitation to study purpose (e.g., intervention, injury prediction)	Systematic and other reviews, editorials, commentaries, case reports or other secondary research
Active duty military populations	Veteran military populations
Published manuscripts or technical reports from 2015 in PubMed, Scopus, CINAHL, SPORTdiscus or Defense Technical Information Center (DTIC) databases	Letters, editorials, comments, abstracts, conference presentations, or book chapters
English-language	

Table 1. Inclusion and Exclusion Criteria

For each of the full texts of included papers, two authors (JAN, CLM, SWM or EMW) will independently extract information into COVIDENCE using the extraction template in Appendix B. Specifically, we will identify the reported study design, population and setting, the case definition(s) of MSKI, whether incident/reinjury/exacerbation MSKI were estimated separately, if prior injury history was measured, whether an estimate of time-loss was reported, and the data sources for each of these outcomes. After the first ten articles have been independently abstracted, the data will be compared and discussed with EMW to ensure agreement and to determine if any additional variables should be included for abstraction. No critical appraisal or quality assessment of the included articles will be performed. The final findings will be summarized in tables and figures as appropriate to depict the extent, range and nature of the queried outcomes as defined in the included articles (Appendix C).

This scoping review follows the framework for systematic scoping reviews published by the Joanna Briggs Institute and incorporates recommendations from other previously published guidelines [34,36,37]. Additional instruction on scoping reviews, including a guidance document and template, and a scoping review registration protocol published on the Open Science Framework were referenced in the creation of this protocol [38–40]. Results of this review will be reported in accordance with the PRISMA-ScR guidelines[41]. Any changes to this review protocol will be documented in the methods section of the manuscript prepared for publication.

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Appendix A

Sample PubMed search conducted on April 17, 2025.

Search	Query	Number of Results
1	("ankle injuries"[TW] OR "ankle injury"[TW] OR "arm injuries"[TW] OR "arm injury"[TW] OR "athletic injuries"[TW] OR "athletic injury"[TW] OR "back injuries"[TW] OR "back injury"[TW] OR "back pain"[TW] OR "bone injuries"[TW] OR "bone injury"[TW] OR "bone pain"[TW] OR "bone pains"[TW] OR dislocate[TW] OR dislocated[TW] OR dislocates[TW] OR dislocation[TW] OR "forearm injuries"[TW] OR "forearm injury"[TW] OR fracture[TW] OR fractured[TW] OR fractures[TW] OR "groin injuries"[TW] OR "groin injury"[TW] OR "hand injuries"[TW] OR "hand injury"[TW] OR "hip injuries"[TW] OR "hip injury"[TW] OR "joint injuries"[TW] OR "joint injury"[TW] OR "joint pain"[TW] OR "joint pains"[TW] OR "knee injuries"[TW] OR "knee injury"[TW] OR "knee pain"[TW] OR "labral tear"[TW] OR "labral tears"[TW] OR "leg injuries"[TW] OR "leg injury"[TW] OR "ligament injuries"[TW] OR "ligament injury"[TW] OR "ligament rupture"[TW] OR "ligament ruptures"[TW] OR "ligament tear"[TW] OR "ligament tears"[TW] OR "lumbar injuries"[TW] OR "lumbar injury"[TW] OR "meniscal tear"[TW] OR "meniscal tears"[TW] OR "muscle injuries"[TW] OR "muscle injury"[TW] OR "muscle strain"[TW] OR "muscle strains"[TW] OR "muscle tear"[TW] OR "muscle tears"[TW] OR "musculoskeletal injuries"[TW] OR "musculoskeletal injury"[TW] OR "musculoskeletal pain"[TW] OR "musculoskeletal pains"[TW] OR "neck injuries"[TW] OR "neck injury"[TW] OR	831,182

Search	Query	Number of Results
	"repetitive strain"[TW] OR "repetitive stress"[TW] OR "rotator cuff injuries"[TW] OR "rotator cuff injury"[TW] OR "shoulder injuries"[TW] OR "shoulder injury"[TW] OR "spinal cord injuries"[TW] OR "spinal cord injury"[TW] OR "spinal injuries"[TW] OR "spinal injury"[TW] OR sprain[TW] OR sprains[TW] OR "stress fracture"[TW] OR "stress fractures"[TW] OR "stress injuries"[TW] OR "stress injury"[TW] OR tendinopathy[TW] OR "tendon injuries"[TW] OR "tendon injury"[TW] OR "tendon pain"[TW] OR "tendon pains"[TW] OR tendonitis[TW] OR "thigh injuries"[TW] OR "thigh injury"[TW] OR "torn ligament"[TW] OR "torn ligaments"[TW] OR "torn meniscus"[TW] OR "torn muscle"[TW] OR "torn muscles"[TW] OR "torn tendon"[TW] OR "torn tendons"[TW] OR "wrist injuries"[TW] OR "wrist injury"[TW] OR "arm injuries"[MeSH] OR "athletic injuries"[MeSH] OR "back injuries"[MeSH] OR "Back pain"[MeSH] OR "fractures, bone"[MeSH] OR "Fractures, Cartilage"[MeSH] OR "fractures, stress"[MeSH] OR "hand injuries"[MeSH] OR "Hip injuries"[MeSH] OR "Joint dislocations"[MeSH] OR "leg injuries"[MeSH] OR "Ligaments/injuries"[MeSH] OR "Musculoskeletal Pain"[MeSH] OR "Musculoskeletal System/injuries"[MeSH] OR "Neck injuries"[MeSH] OR "Neck pain"[MeSH] OR "Occupational injuries"[MeSH] OR "Rib Fractures"[MeSH] OR "Rotator Cuff injuries"[MeSH] OR "Shoulder injuries"[MeSH] OR "Shoulder pain"[MeSH] OR "Spinal cord injuries"[MeSH] OR "spinal injuries"[MeSH] OR "Sprains and Strains"[MeSH] OR "Tendon injuries"[MeSH])	

Search	Query	Number of Results
2	("active duty"[TW] OR "air force"[TW] OR airforce[TW] OR airman[TW] OR airmen[TW] OR "armed forces"[TW] OR armies[TW] OR army[TW] OR cadet[TW] OR cadets[TW] OR "coast guard"[TW] OR corpsman[TW] OR corpsmen[TW] OR defence force[TW] OR "defence forces"[TW] OR defense force[TW] OR "defense forces"[TW] OR guardian[TW] OR infantry[TW] OR "Marine corps"[TW] OR marines[TW] OR midshipman[TW] OR midshipmen[TW] OR militaries[TW] OR military[TW] OR "national guard"[TW] OR naval[TW] OR navies[TW] OR navy[TW] OR officer[TW] OR parachuting[TW] OR parachutist[TW] OR parachutists[TW] OR paratrooper[TW] OR paratroopers[TW] OR reservist[TW] OR reservists[TW] OR sailor[TW] OR sailors[TW] OR "service member"[TW] OR "service members"[TW] OR servicemember[TW] OR servicemembers[TW] OR servicemen[TW] OR servicewomen[TW] OR soldier[TW] OR soldiers[TW] OR "space force"[TW] OR "special forces"[TW] OR "special operations"[TW] OR troops[TW] OR "uniformed services "[TW] OR "Military Health"[MeSH] OR "Military Personnel"[MeSH])	148,144
3	#1 AND #2	7,398
4	#3 NOT ("oceans and seas"[mesh])	7,385
5	#4 NOT (pediatric[ti] OR paediatric[ti])	7,352

Search	Query	Number of Results
6	#5 NOT ("case reports"[pt] OR comment[pt] OR editorial[pt] OR letter[pt] OR "review"[pt] OR "systematic review"[pt] OR "scoping review"[pt] OR "scoping review"[ti] OR "systematic review"[ti])	5,637
7	#6 NOT (animals[mesh] NOT humans[mesh])	5,519
8	#7 AND 2014:2030[pdat]	2,694

Appendix B

General information

Study ID

Title

Full title of paper/report the data are extracted from

Author, year

Please follow the following format:

Three or more authors:

Smith et al., 2024

Two authors:

Smith & Stannard, 2024

One author:

Smith, 2024

Lead author email

If published

Nation of military population studied:

Select country of military population studied.

If "Mixed/Multinational" or "other", please write country(s) in.

1. United States
2. UK
3. Canada
4. Australia
5. Mixed/Multinational
6. Other

Study Setting

Basic / Initial Military Training means the first military training environment when one joins.

The specific name may vary by branch or country (i.e., boot camp, BCT, IET, AIT, etc.)

Garrison setting is the normal non-deployed military working and training environment.

Training outside of initial military training could include things like Ranger school, Special Forces Assessment and Selection, and other advanced training schools.

Deployed environment is away from garrison for an operational reason (i.e. combat, humanitarian or peacekeeping mission, etc.)

If you do not feel the study is contained in any of those settings please select other and add "free text".

1. Basic / Initial Military Training
2. Garrison
3. Training outside of initial military training (i.e., Ranger school)
4. Deployed environment
5. Other

Study Population Type

Select the type of individuals comprising the study population.

Check all that apply.

1. Trainees
2. Non-Trainee, regular military
3. Special operations (e.g., Special Forces, Ranger, SEAL)
4. Not specified
5. Other

Study design

Select the study design as reported by the authors.

If other, please populate free text.

1. Retrospective cohort study

2. Prospective cohort study
3. Cross-sectional study
4. Case-control study
5. Randomized controlled trial
6. Cluster randomized controlled trial
7. Interrupted Time Series / Controlled Interrupted Time Series
8. Other

Study follow-up period

Please write number of months individuals were followed for.

Example: 12 weeks of follow-up = 3

180 days of follow-up = 6

Musculoskeletal Injury (MSKI) Case Definitions

An MSKI case definition was provided.

1. Yes
2. No

MSKI cases were defined as injuries requiring medical attention.

1. Yes
2. No

MSKI cases were defined as injuries requiring time-loss.

If time-loss was required to be an MSKI case, this answer should be "yes".

If a case definition required both medical attention and time-loss, these both would be "yes".

If time-loss injuries were considered a subset of medical-attention injury cases, this question should be "no", as the actual case was based on medical attention.

1. Yes
2. No

MSKI cases were defined as all complaints, regardless of medical attention or time-loss status.

If this were the case a mechanism to capture non-medical attention injuries would be required (e.g., survey or interview).

1. Yes
2. No

Was MSKI case defined in another way? If so, specify in the subsequent notes section.

1. Yes
2. No

If another specified MSKI case definition was used please indicate here.

Examples:

Medical attention overuse injuries.

Time-loss injuries that resulted in recycling training (having to restart basic training, etc.).

Was more than one case definition used?

This should be "yes" only if multiple discrete definitions / analyses were used.

i.e., medical attention injuries from EMR as one case definition and analyses; time-loss injuries from an admin data source as another;

If a single case definition required both time-loss and medical attention, the answer would be "NO".

1. Yes
2. No

If multiple case definitions were used, please indicate here. :

Example:

Medical attention; Time-loss

What data source(s) were used to identify MSKI cases?

If another source was used please specify in the free text box.

1. Electronic Medical Records (EMR)
2. Clinician report outside of EMR (e.g., injury tracker)
3. Non-clinical administrative system (e.g., cadre managed, limited duty day tracker, etc.)
4. Individual self-report including survey
5. Not specified
6. Other

MSKI Outcomes and Estimates Reported

The purpose of these next questions is to identify WHAT made up the N for estimates of MSKI occurrence.

Was a separate estimate for incident MSKI occurrence presented?

1. Yes
2. No

Was a separate estimate for reinjury MSKI occurrence presented?

Reinjury is defined as an injury of the same type and to the same body region from which the individual HAD PREVIOUSLY recovered.

1. Yes
2. No

Was a separate estimate for exacerbation MSKI occurrence presented?

Exacerbation is defined as an injury of the same type and to the same body region from which the individual HAD NOT PREVIOUSLY recovered.

1. Yes
2. No

If separate estimates were not presented, which injuries comprised the estimate(s); Select only what is known. If it is known that new injuries were included, but it is not clear whether reinjuries or exacerbations were included, select:

-New or Incident

1. New or Incident
2. Reinjury
3. Exacerbation

What estimate(s) of MSKI were reported:

1. Incidence (i.e., risk)
2. Incidence Rate
3. Incidence Rate Ratio
4. Hazard Ratio
5. Odds Ratio
6. Prevalence
7. Other

Previous Injury History

Was a previous history of injury measured?

1. Yes
2. No

What were the data sources for obtaining previous history of injury?

1. Electronic Medical Records (EMR)
2. Clinician report outside of EMR (e.g., injury tracker)
3. Non-clinical administrative system (e.g., cadre managed, limited duty day tracker, etc.)
4. Individual self-report including survey
5. Not specified
6. Other

Time-loss

Was an estimate of time-loss reported?

1. Yes
2. No

How was time-loss defined?

If time-loss was defined in a manner other than a limited duty day (i.e., restricted from full military occupational requirements and/or physical training), please select other and specify in the field.

1. Limited duty day (e.g., a day in which individual was unable to fully perform military requirements)
2. Other

How was time-loss aggregated and reported?

If reported in another way please annotate.

1. Time-loss per person-time
2. Time-loss per injury
3. Other

What was the source(s) for time-loss data?

1. Electronic Medical Records (EMR)
2. Clinician report outside of EMR (e.g., injury tracker)
3. Non-clinical administrative system (e.g., cadre managed, limited duty day tracker, etc.)
4. Individual self-report including survey
5. Not specified
6. Other

Internal Reviewer Notes

Additional comments for review team (i.e., good article examples to highlight, additional fields we should add, etc.). For articles to highlight, be looking for articles that had defined case definition and included separate estimates of MSKI (i.e. incident and reinjury).

Appendix C

Sample Table Shells

Author, year	Nationality	Setting	Population Type	Study Design	Length of Follow-up	Musculoskeletal Injury Case Definitions				
						All complaints	Medical Attention	Time-loss	Other	Data Source

Sample Table 1. Study Overview and Musculoskeletal Injury Case Definitions

Musculoskeletal Injury Occurrence Estimates						Prior Injury History		Time-loss Injury		
Separate estimate for incident MSKI?	Separate estimate for reinjury MSKI?	Separate estimate for exacerbation MSKI?	Combined estimate that included:			Previous injury history measured?	Data Source for Previous Injury History:	Estimate of Time-loss reported?	How was time-loss aggregated and reported?	Time-loss data source:
			Incident MSKI	Reinjury MSKI	Exacerbation MSKI					

Sample Table 2. Musculoskeletal Injury Occurrence Estimates, Prior Injury History and Time-loss Injury Variables